

Aircraft Gas Turbine Engine

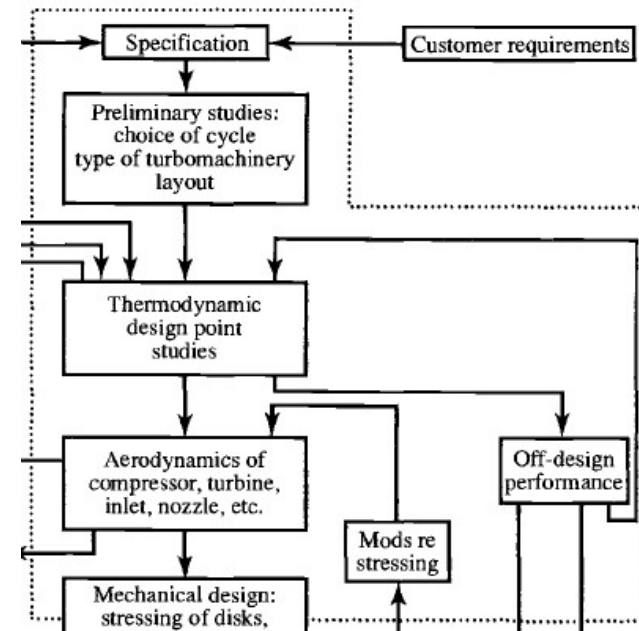
Class 4 – Off – Design Cycle Analysis

Introduction

- The parametric cycle analysis described in the past lecture showed how design point engine and component behavior was determined by design choices.
- The object of off design point analysis is to determine the engine's performance over its operating envelope. The performance of several different engines can then be compared to reveal trends in engine performance with design choices.

Off Design Engine Concept

- Once an engine cycle and its design choices are made, a candidate engine is sized/scaled in terms of mass flow rate based on sea level static (SLS) thrust which is known as parametric cycle analysis (design point).
- In off design, the design choices have been made, and the performance of this specific reference point engine is needed at all possible operating conditions. *The independent variables in for this scenario are flight conditions, throttle settings and nozzle settings.*



Mass Flow Parameter

- The mass flow parameter (MFP) or non dimensional mass flow is used to capitalize on the fundamental law of conservation of mass.

$$NDMF = MFP = \frac{\dot{m}\sqrt{T_t}}{P_t A} = M \sqrt{\frac{\gamma}{R}} \left(\frac{1}{1 + \frac{\gamma-1}{2} M^2} \right)^{\frac{\gamma+1}{2(\gamma-1)}}$$

- The MFP relates the cross-sectional area and Mach number at any engine station. It represents the corrected mass flow rate per unit area and has a maximum value at $M = 1$. Thus, for a given total temperature and total pressure, the maximum mass flow rate per area corresponds to a flow Mach number of unity.

Engine Corrected Parameters

- There are five component and engine corrected parameters. These are:
 - 1) Corrected Mass Flow Rate
 - 2) Corrected Shaft Speed
 - 3) Corrected Fuel Flow Rate
 - 4) Corrected Thrust
 - 5) Corrected Thrust Specific fuel consumption

Engine Corrected Parameters

- Corrected mass flow rate and corrected shaft speed are the parameters that fully describe the behavior of a compressor or turbine. A reference pressure and temperature, $p_{ref} = 101.33 \text{ kPa}$ and $T_{ref} = 288.2 \text{ K}$, should be defined in order to define a correction parameter, for station i :

$$\theta_i \equiv \frac{T_{ti}}{T_{ref}}$$

$$\delta_i \equiv \frac{p_{ti}}{p_{ref}}$$

Engine Corrected Parameters

- Corrected mass flow rate at station I is the mass flow rate through station i at the reference pressure and temperature:

$$\dot{m}_{ci} = \frac{\dot{m}\sqrt{\theta_i}}{\delta_i} = \frac{p_{ref}A_i}{\sqrt{T_{ref}}} MFP_i$$

- Thus, the corrected mass flow rate per unit area at any engine station is easily determined using the Mach number at that station, since MFP_i is a function of M_i for known gas properties.

Engine Corrected Parameters

Corrected Shaft Speed

$$N_{ci} \equiv \frac{N}{\sqrt{\theta_i}}$$

Corrected fuel flow rate

$$\dot{m}_{fc} \equiv \frac{\dot{m}_f}{\delta_2 \sqrt{\theta_2}}$$

Corrected thrust

$$F_c \equiv \frac{F}{\delta_0}$$

Corrected thrust specific fuel consumption

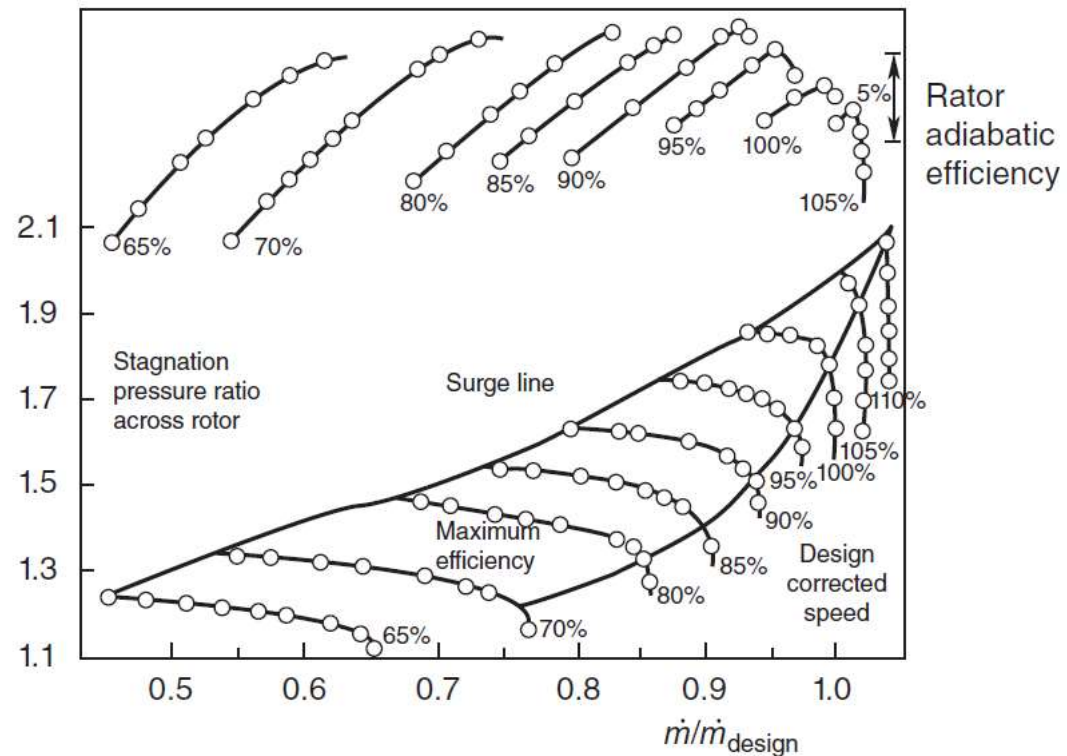
$$TSFC_c \equiv \frac{TSFC}{\sqrt{\theta_2}}$$

By combining previous equations, it results in

$$TSFC_c \equiv \pi_d \frac{\dot{m}_{fc}}{F_c}$$

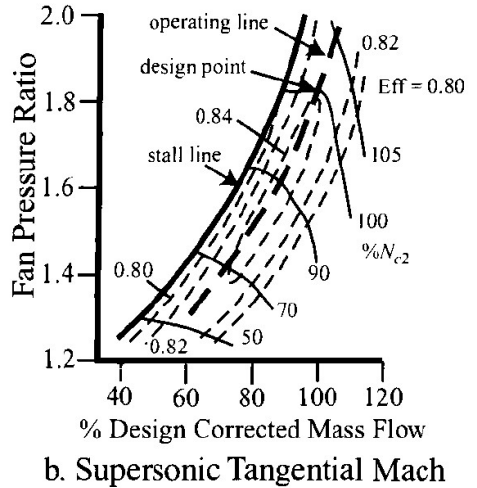
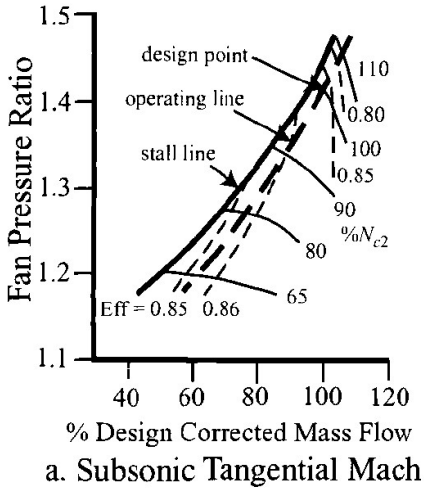
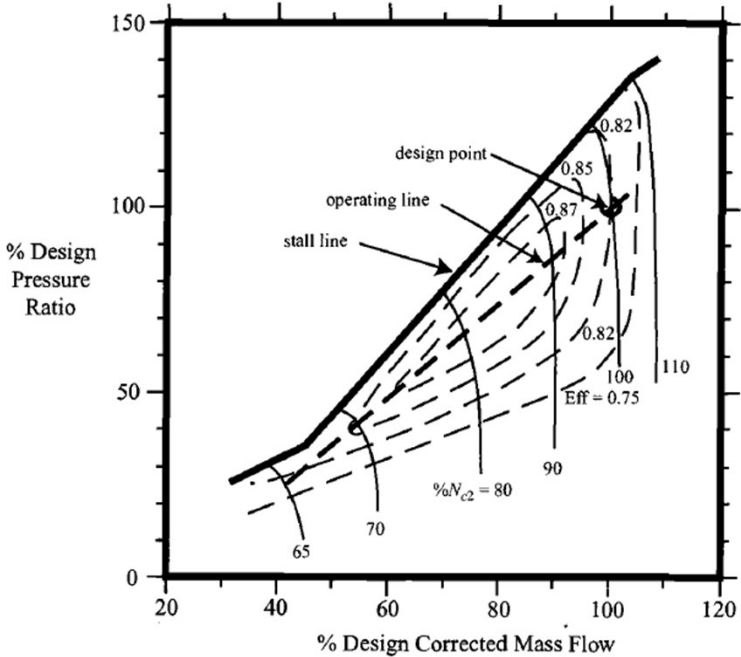
Fan and Compressor Maps

- The performance map of a compressor or fan is normally presented using the following performance parameters:
 - Total Pressure Ratio
 - Corrected Mass Flow Rate
 - Corrected Engine Speed
 - Adiabatic Efficiency
- Variations in the axial flow velocity in response to changes in pressure cause the multistage compressor to have quite different mass flow vs pressure ratio characteristics than one of its stages



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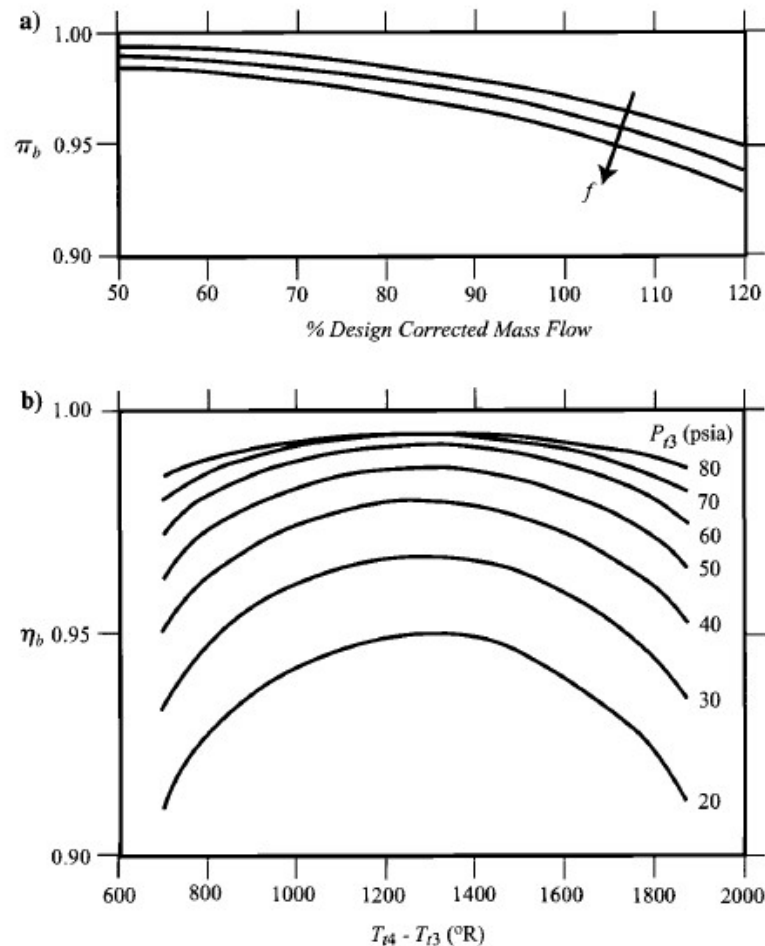
Examples



Combustor Maps

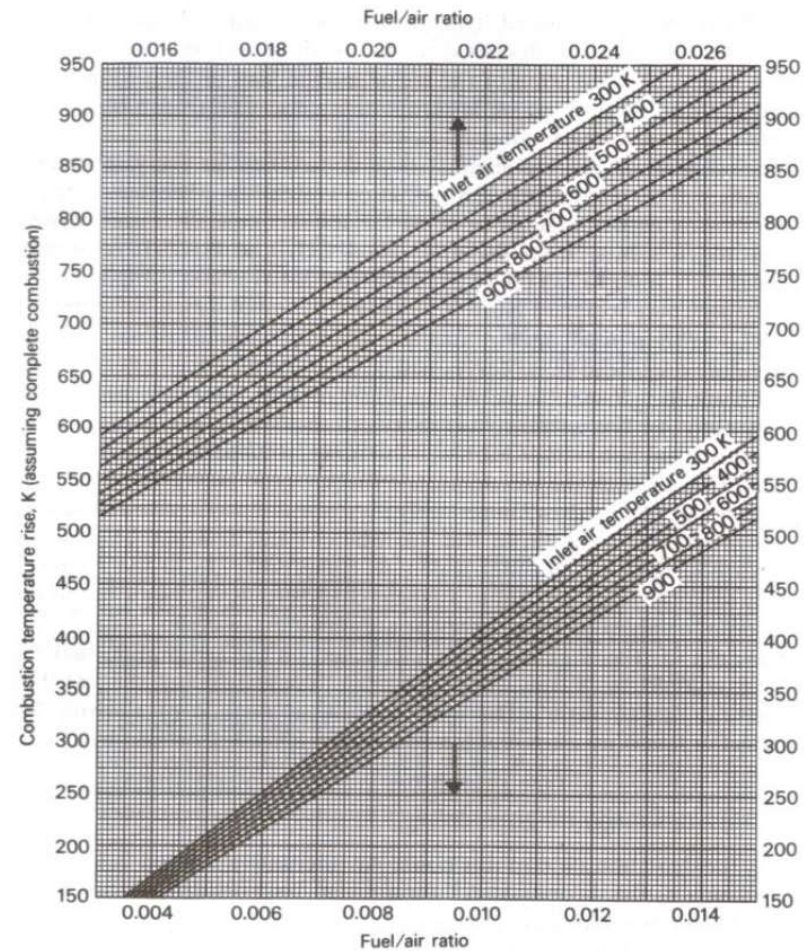
- The combustor performance that are most important to overall engine performance are the total pressure loss through the combustor and combustion efficiency.

- The pressure loss of a combustor is normally plotted vs the corrected mass flow rate for different fuel/air ratios.
- The efficiency of the combustor is plotted vs the temperature rise in the combustor or fuel/air ratios for various values of inlet pressure.



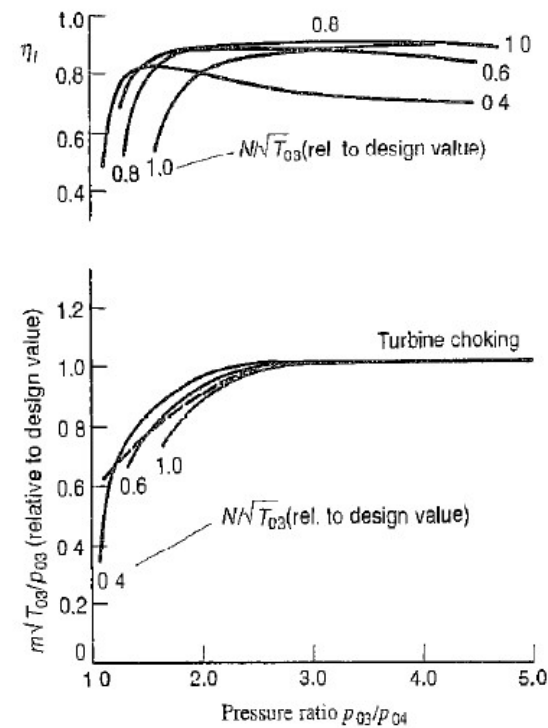
Combustor Maps

- Example of temperature rise combustor map, obtained via polynomials and fuel to air ratio iterations.



Turbine Maps

- The performance of a turbomachinery stage is fully determined by two parameters; (1) the axial Mach number or equivalently the corrected mass flow rate and (2) the tangential blade Mach number, or equivalently the corrected shaft speed. The turbine performance map is thus a graph of π versus the corrected mass flow rate $\dot{m}_{c4}$ for the corrected shaft speeds N_{c4} .



Engine Off-design Performance Modelling

- To estimate the performance of any engine at a variety of operating and ambient conditions we must model the thermodynamic phenomena. This model is simply a “joining together” of the component maps. The normal laws of physics are then used to obtain the estimated performance at any desired condition:
 - Mass flow continuity
 - Conservation of energy
 - Compressor and turbines on the same shaft rotate at the same speed
 - Gas Laws
- Mathematical procedures are used to make each chosen operating point “converge” to a correct solution.

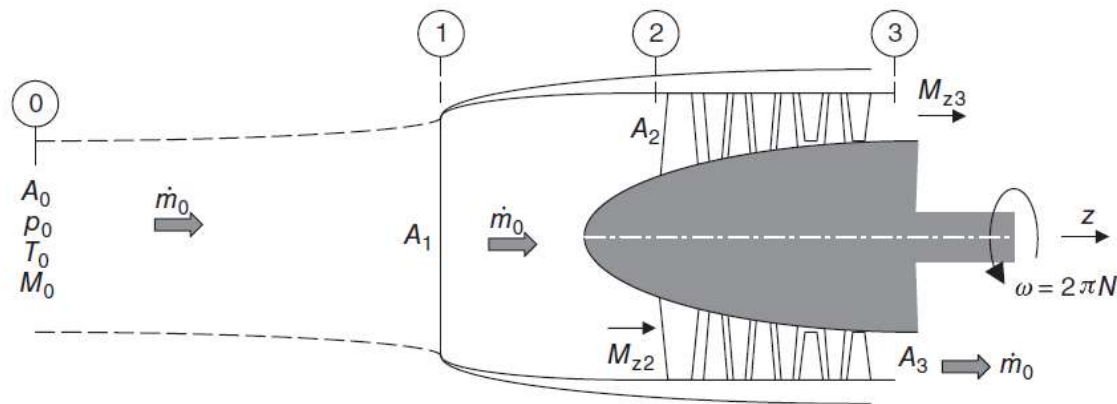
Fundamental Assumptions

1. The flow areas are constants at *turbine inlet and outlet both high and low pressures. Afterburner inlet and engine nozzle throat (when afterburner off).*
2. *The flow is choked at the high – pressure turbine entrance nozzles and at the low – pressure turbine entrance nozzles.*
3. Component efficiencies (fan, compressor, burner, turbine , afterburner and mechanical) and total pressure ratios (burner, mixer, afterburner and nozzle) *do not change from their design values.*
4. Bleed air and cooling air fractions are constant. Power take off constant.
5. Fuel – air ratios are ignored when compared to unity.
6. The exit area A_9 of the exhaust nozzle is adjustable so that the pressure ratio P_0/P_9 can be set to a predetermined value.

Component Matching

- Inlet – Compressor Matching

$$\pi_d \cdot \frac{A_2}{A_0} \cdot M_{z2} \left(1 + \frac{\gamma - 1}{2} M_{z2}^2 \right)^{\frac{-(\gamma+1)}{2(\gamma-1)}} = M_0 \left(1 + \frac{\gamma - 1}{2} M_0^2 \right)^{\frac{-(\gamma+1)}{2(\gamma-1)}}$$



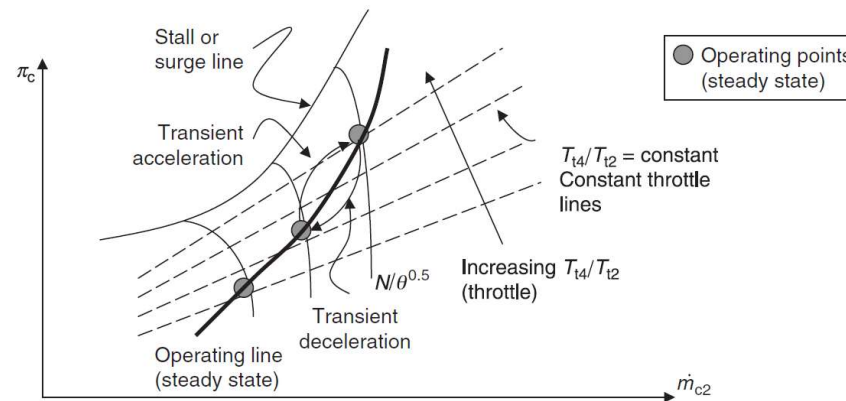
Component Matching

- Compressor – Combustor Matching
 - Mass balance

Remember: $\dot{m}_4 = \dot{m}_3 + \dot{m}_f = (1 + f)\dot{m}_2$

Combustor Inlet (compressor discharge) = Combustor Outlet (Turbine Inlet)

$$(1 + f) \sqrt{\frac{\gamma_c}{R_c}} A_2 \cdot M_{z2} \left(1 + \frac{\gamma_c - 1}{2} M_{z2}^2 \right)^{\frac{-(\gamma_c + 1)}{2(\gamma_c - 1)}} = \sqrt{\frac{\gamma_t}{R_t}} \cdot \frac{\pi_b \pi_c}{\sqrt{T_{t4}/T_{t2}}} \cdot A_4 \left(\frac{\gamma_t + 1}{2} \right)^{\frac{-(\gamma_t + 1)}{2(\gamma_t - 1)}}$$



Component Matching

- Compressor – Combustor Matching
 - Energy balance

$$f = \frac{h_{t4} - h_{t3}}{Q_R \eta_b - h_{t4}}$$

- If the throttle setting has been specified $Tt4$ in the off-design condition. Once we establish a compressor pressure ratio at the off-design operation, h_{t3} becomes known. We may use an assumed value for the burner efficiency correlation for η_b to proceed with the calculation of the new fuel-to-air ratio f at off-design.

Component Matching

- Combustor – Turbine Matching
- The steady state combustor – turbine interaction is dominated by:
 - 1) Choked turbine nozzle
 - 2) Turbine cooling requirements
 - 3) Hot spots in turbine entry
 - 4) Loss of turbine efficiency due to cooling
- The main objective in turbine off-design analysis is primarily to calculate turbine expansion parameter.

Component Matching

- Combustor – Turbine Matching
 - The corrected mass flow rates, in the cooled turbine case, are related by cooling fractions

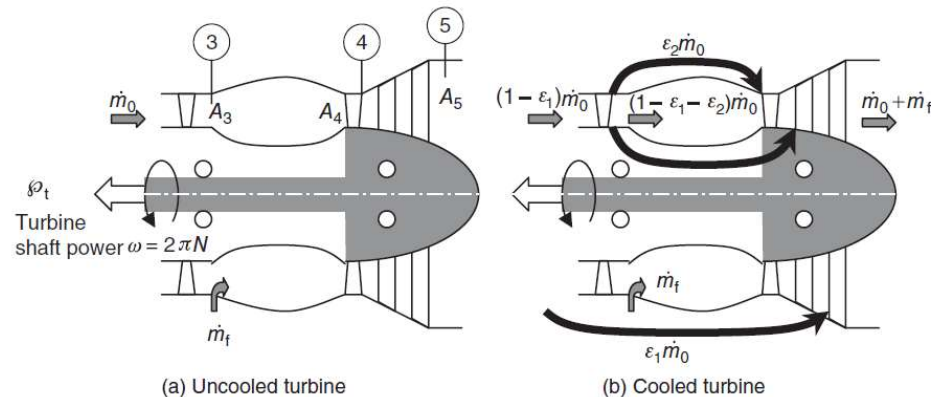
$$\frac{\dot{m}_{c4}}{\dot{m}_{c5}} = \left(1 - \frac{\varepsilon_1 + \varepsilon_2}{1 + f} \right) \frac{\pi_t}{\sqrt{\tau_t}}$$

ε : Cooling fraction for turbine nozzles

where

$$\frac{\dot{m}_{c4}}{\dot{m}_{c5}} = \frac{\dot{m}_4 \sqrt{\theta_4/\delta_4}}{\dot{m}_5 \sqrt{\theta_5/\delta_5}}$$

Reference conditions are the design point conditions for each station.



Component Matching

- Compressor – Turbine Power Matching
 - From the power balance $W_c = W_T \times \eta_m$ the following can be calculated:

$$\frac{T_{t3}}{T_{t2}} - 1 = \eta_m(1 + f) \frac{Cp_t}{Cp_c} \frac{T_{t4}}{T_{t2}} \left(1 - \frac{T_{t5}}{T_{t4}}\right)$$

- Assuming the expansion parameter as constant (valid for a wide operating range in the engine off design conditions), we may conclude that the compression varies linearly with the throttle parameter T_{t4}/T_{t2} in off – design. Therefore:

$$\frac{\tau_{c,OD} - 1}{\tau_{c,D} - 1} \cong \frac{(T_{t4}/T_{t2})_{OD}}{(T_{t4}/T_{t2})_D}$$

Component Matching

- Compressor – Turbine Mass flow Matching
 - From the mass balance $\dot{m}_4 = (1 + f)\dot{m}_2$ the following can be calculated:

$$\dot{m}_{c2} = \frac{\dot{m}_{c4} \frac{p_{t4}}{p_{t2}}}{1 + f} \sqrt{\frac{T_{t4}}{T_{t2}}} = \frac{\dot{m}_4 \pi_b \pi_c}{(1 + f) \sqrt{T_{t4}/T_{t2}}}$$

- Where the flow in station 4 is choked (over a wide operating range) making the corrected mass Flow constant. Therefore:

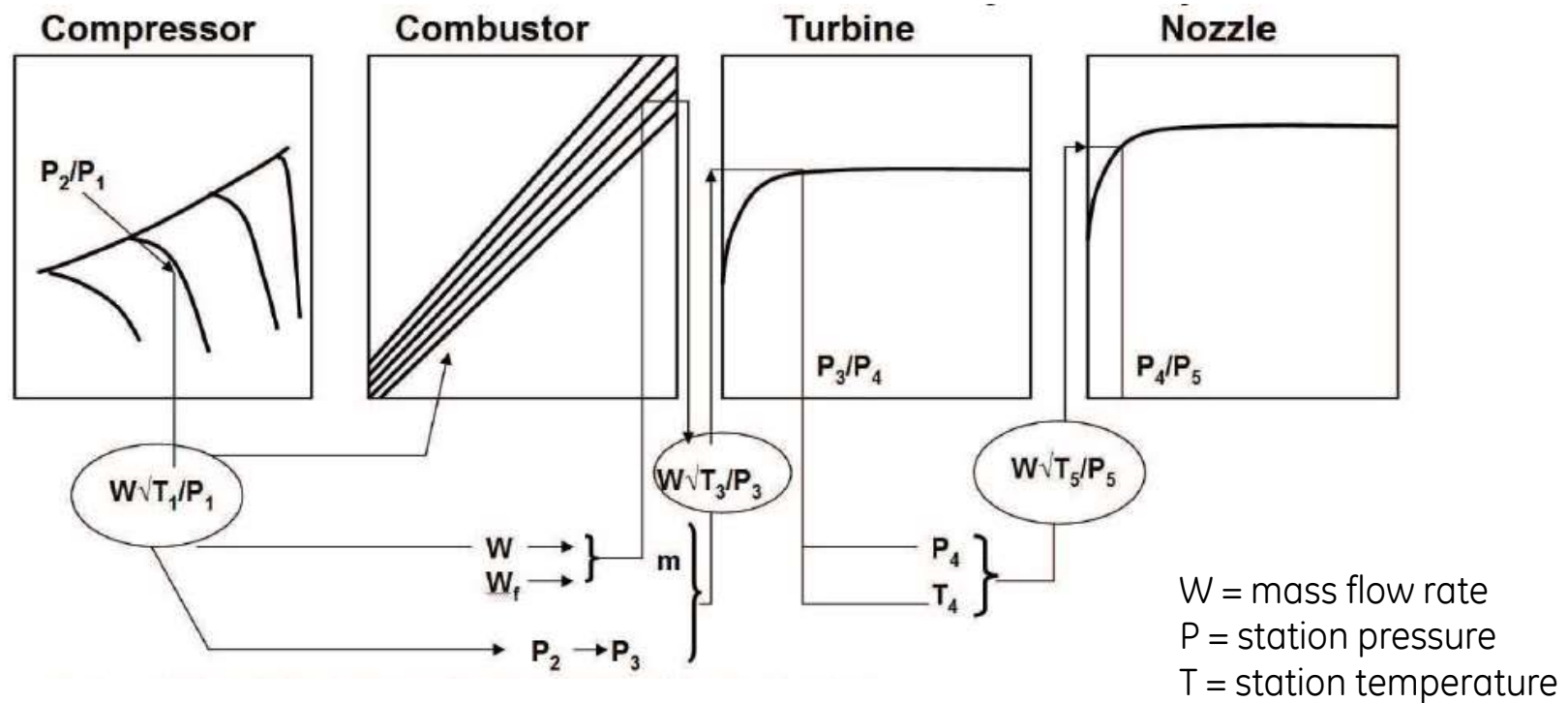
$$\frac{\dot{m}_{c2,OD} - 1}{\dot{m}_{c2,D} - 1} \cong \frac{\pi_{C,OD}}{\pi_{C,D}} \frac{\sqrt{(T_{t4}/T_{t2})_D}}{\sqrt{(T_{t4}/T_{t2})_{OD}}}$$

Component Matching

- Compressor – Turbine Mass flow Matching
 - where the compressor pressure ratio is given by:

$$\pi_{C,OD} = \left[1 + \frac{(T_{t4}/T_{t2})_{OD}}{(T_{t4}/T_{t2})_D} \left(\pi_{C,D}^{(\gamma_c-1)/\gamma_c} - 1 \right) \right]^{\gamma_c/(\gamma_c-1)}$$

Component Matching – Graphic Representation



Engine Off-design Calculation

Iterative Solution Scheme – Cranfield University

Example:

The following data correspond to a two-spool engine turbojet with a convergent nozzle at design point conditions. The engine is later operated in the specified off-design conditions. What is the thrust and SFC for both design and off-design conditions?

Design Point Conditions			
Flight conditions			
Static			
Temperature	C		15
Pressure	kPa		101.325
M0	-		0
Design			
Pressure Ratios			
Inlet	-		0.985
Compressor	-		15
Burner	-		0.95
Nozzle	-		0.98
Efficiencies			
Compressor Isentropic	-		0.85
Burner	-		0.98
Turbine Isentropic			0.85
Shaft/Mechanical			0.995
TIT	K		1480
Air mass flow rate	kg/s		150

Off-design Point Conditions			
Flight conditions			
Static			
Temperature	C		- 63
Pressure	kPa		7
M0	-		1.8
Design			
Pressure Ratios			
Intake Pressure Recovery			0.94
TIT	K		1600
Gas Properties			
Cp Cold Section	J/kg.K		1004
γ cold	-		1.4
Cp Hot Section	J/kg.K		1156
γ hot	-		1.33
FHV	kJ/kg		42.000

Inlet

Design Point

$$T_{t0} = 288 \text{ K}$$

$$P_{t0} = 101.3 \text{ kPa}$$

$$a_0 = 340.09 \text{ m/s} ; V_0 = 0 \text{ m/s};$$

$$P_{t2} = P_{t0} \pi_d = 101.3 \cdot 0.985 \\ = 99.81 \text{ kPa}$$

The analysis for the off - design starts just as the design conditions, assuming an adiabatic inlet.

The inlet discharge pressure is the pressure times the pressure ratio ($P_{\text{exit}}/P_{\text{inlet}}$).

Off Design

$$T_{t0} = 346.33 \text{ K}$$

$$P_{t0} = 40.22 \text{ kPa}$$

$$a_0 = 289.83 \text{ m/s} ; V_0 \\ = 521.88 \text{ m/s};$$

$$P_{t2} = P_{t0} \pi_d = 40.22 \text{ kPa} \cdot 0.94 \\ = 37.81 \text{ kPa}$$

LP Compressor

Compressor Inlet Conditions:

The iterative solution scheme relies on the power balance based on the turbine expansion. Therefore one of the first guesses is the **compressor pressure ratio**.

Followed by the step-by-step procedure:

Design Point

$$T_{t2} = 288.15 \text{ K}$$

$$P_{t2} = 101.325 \text{ kPa}$$

$$P_{t2.5} = P_{t2} \pi = 99.81 \text{ kPa} \cdot 3.87 = 386.54 \text{ kPa}$$

$$T_{t2.5 \text{ ideal}} = T_{t2} \left(\frac{P_{t2.5}}{P_{t2}} \right)^{\frac{\gamma-1}{\gamma}} = 424.26 \text{ K}$$

Design Point

$$\eta_{is} = \frac{T_{t2.5 \text{ ideal}} - T_{t2}}{T_{t2.5} - T_{t2}} \left(\frac{\text{Ideal}}{\text{Real}} \right) \Rightarrow T_{t2.5} = T_{t2} + \frac{T_{t2.5 \text{ ideal}} - T_{t2}}{\eta_{is}} = 288 + \frac{424.26 - 288}{0.85} = 448.28 \text{ K}$$

$$W_{CLP} = \dot{m}_{air} C_{p,c} (T_{t2.5} - T_{t2}) = 24.115 \text{ MW}$$

Off-design Point

$$\eta_{is} = \frac{T_{t2.5 \text{ ideal}} - T_{t2}}{T_{t2.5} - T_{t2}} \left(\frac{\text{Ideal}}{\text{Real}} \right) \Rightarrow T_{t2.5} = T_{t2} + \frac{T_{t2.5 \text{ ideal}} - T_{t2}}{\eta_{is}} = 346.08 + \frac{493.46 - 346.33}{0.85} = 519.43 \text{ K}$$

$$W_{CLP} = \dot{m}_{air} C_{p,c} (T_{t2.5} - T_{t2}) = 7.806 \text{ MW}$$

Off Design

$$T_{t2} = 346.33 \text{ K}$$

$$P_{t2} = 37.81 \text{ kPa}$$

$$P_{t2.5} = P_{t2} \pi = 37.81 \text{ kPa} \cdot \mathbf{3.4529} = 130.54 \text{ kPa}$$

$$T_{t2.5 \text{ ideal}} = T_{t2} \left(\frac{P_{t2.5}}{P_{t2}} \right)^{\frac{\gamma-1}{\gamma}} = 493.46 \text{ K}$$

Smart guess

HP Compressor

As in the low-pressure compressor the same procedure should be followed. A new pressure ratio is assumed as well and later the expansion should be confirmed in the HP turbine.

Design Point

$$T_{t2.5} = 448.28 \text{ K}$$

$$P_{t2.5} = 386.54 \text{ kPa}$$

$$P_{t3} = P_{t2.5} \pi = 1497.08 \text{ kPa}$$

$$T_{t3 \text{ ideal}} = T_{t2.5} \left(\frac{P_{t3}}{P_{t2.5}} \right)^{\frac{\gamma-1}{\gamma}} = 660.03 \text{ K}$$

Design Point

$$\eta_{is} = \frac{T_{t3 \text{ ideal}} - T_{t2.5}}{T_{t3} - T_{t2.5}} \Rightarrow T_3 = T_{t2.5} + \frac{T_{t3 \text{ ideal}} - T_{t2.5}}{\eta_{is}} = 697.39 \text{ K}$$

$$W_{\text{CHP}} = \dot{m}_{\text{aire}} C_p (T_{t3} - T_{t2.5}) = 37.516 \text{ MW}$$

Off-design Point

$$\eta_{is} = \frac{T_{t3 \text{ ideal}} - T_{t2.5}}{T_{t3} - T_{t2.5}} \Rightarrow T_3 = T_{t2.5} + \frac{T_{t3 \text{ ideal}} - T_{t2.5}}{\eta_{is}} = 788.70 \text{ K}$$

$$W_{\text{CHP}} = \dot{m}_{\text{aire}} C_p (T_{t3} - T_{t2.5}) = 12.144 \text{ MW}$$

Off Design

$$T_{t2.5} = 519.43 \text{ K}$$

$$P_{t2.5} = 130.54 \text{ kPa}$$

$$P_{t3} = P_{t2.5} \pi = 130.59 \text{ kPa} \cdot \mathbf{3.5889} = 468.51 \text{ kPa}$$

$$T_{t3 \text{ ideal}} = T_{t2.5} \left(\frac{P_{t3}}{P_{t2.5}} \right)^{\frac{\gamma-1}{\gamma}} = 748.31 \text{ K}$$

Smart guess

Combustor

Design point

Combustor efficiency is a given value. Therefore the fuel mass flow can be calculated.

$$\eta_{cc} = \frac{(\dot{m}_0 + \dot{m}_f)h_4 - \dot{m}_{aire}h_3}{\dot{m}_f(LHV)} \left(\frac{\Delta h_{real}}{\Delta h_{ideal}} \right)$$

$$\Rightarrow \dot{m}_f = \frac{\dot{m}_0(h_4 - h_3)}{(\eta_{cc} \cdot LHV - h_4)} = \frac{150 (1710.88 - 700.18)}{(0.98 \cdot 42000 - 170.88)} = 3.84 \frac{\text{kg}}{\text{s}}$$

Off-design point

Considering we assumed the Flow to be choked at the turbine inlet. An air mass flow rate should be assumed in order to converge with this supposition. Mass flow parameter is used as a check.

$$\eta_{cc} = \frac{(\dot{m}_0 + \dot{m}_f)h_4 - \dot{m}_0h_3}{\dot{m}_f(LHV)} \left(\frac{\Delta h_{real}}{\Delta h_{ideal}} \right)$$

Smart guess

$$\Rightarrow \dot{m}_f = \frac{\dot{m}_0(h_4 - h_3)}{(\eta_{cc} \cdot LHV - h_4)} = \frac{45.10 (1840 - 788.24)}{(0.98 \cdot 42000 - 1840)} = 1.21 \frac{\text{kg}}{\text{s}}$$

$$\left(\frac{\dot{m}_4 \sqrt{T_{t4}}}{P_{t4}} \right)_{DP} = 4.161$$

$$\left(\frac{\dot{m}_4 \sqrt{T_{t4}}}{P_{t4}} \right)_{O-D} = 4.161$$

$$\left(\frac{\dot{m}_4 \sqrt{T_{t4}}}{P_{t4}} \right)_{ERROR} = \frac{\left(\frac{\dot{m}_4 \sqrt{T_{t4}}}{P_{t4}} \right)_{O-D} - \left(\frac{\dot{m}_4 \sqrt{T_{t4}}}{P_{t4}} \right)_{DP}}{\left(\frac{\dot{m}_4 \sqrt{T_{t4}}}{P_{t4}} \right)_{DP}} = 0.00\%$$

HP Turbine

Turbine inlet conditions are given from design characteristics:

Design Point

$$T_{t4} = 1480 \text{ K}$$

$$P_{t4} = 1422.22 \text{ kPa}$$

Off Design

$$T_{t4} = 1600 \text{ K}$$

$$P_{t4} = 445.08 \text{ kPa}$$

From power balance we obtain temperature at turbine exit:

$$T_{t4.5} = T_{t4} - \frac{W_{\text{CHP}}}{\eta_M \dot{m}_H C_{pH}} = 1267.99 \text{ K}$$

$$T_{t4.5} = T_{t4} - \frac{W_{\text{CHP}}}{\eta_M \dot{m}_H C_{pH}} = 1370.80 \text{ K}$$

Followed by the step-by-step procedure:

$$T_{t4.5 \text{ ideal}} = T_{t4} - \frac{T_{t4} - T_{t4.5}}{\eta_{is}} = 1230.57 \text{ K}$$

$$T_{t4.5 \text{ ideal}} = T_{t4} - \frac{T_{t4} - T_{t4.5}}{\eta_{is}} = 1330.35 \text{ K}$$

Design Point

$$\frac{T_{t4.5 \text{ ideal}}}{T_{t4}} = \left(\frac{P_{t4.5}}{P_{t4}} \right)^{\frac{(\gamma-1)}{\gamma}} \Rightarrow P_{t4.5} = P_{t4} \left(\frac{T_{t4.5 \text{ ideal}}}{T_{t4}} \right)^{\frac{\gamma}{\gamma-1}} = 675.96 \text{ kPa}$$

Off-design Point

$$\frac{T_{t4.5 \text{ ideal}}}{T_{t4}} = \left(\frac{P_{t4.5}}{P_{t4}} \right)^{\frac{(\gamma-1)}{\gamma}} \Rightarrow P_{t4.5} = P_{t4} \left(\frac{T_{t4.5 \text{ ideal}}}{T_{t4}} \right)^{\frac{\gamma}{\gamma-1}} = 211.54 \text{ kPa}$$

LP Turbine

Followed by the step-by-step procedure:

Design Point

$$T_{t4.5} = 1267.99 \text{ K}$$

$$P_{t4.5} = 675.96 \text{ kPa}$$

$$T_{t5} = T_{t4.5} - \frac{W_{\text{CHP}}}{\eta_M \dot{m}_H C_{pH}} = 1131.70 \text{ K}$$

$$T_{t5 \text{ ideal}} = T_{t4} - \frac{T_{t4.5} - T_{t5}}{\eta_{\text{is}}} = 1107.65 \text{ K}$$

Off Design

$$T_{t4.5} = 1370.80 \text{ K}$$

$$P_{t4.5} = 211.54 \text{ kPa}$$

$$T_{t5} = T_{t4.5} - \frac{W_{\text{CHP}}}{\eta_M \dot{m}_H C_{pH}} = 1223.46 \text{ K}$$

$$T_{t5 \text{ ideal}} = T_{t4} - \frac{T_{t4.5} - T_{t5}}{\eta_{\text{is}}} = 1197.46 \text{ K}$$

Design Point

$$\frac{T_{t5 \text{ ideal}}}{T_{t4.5}} = \left(\frac{P_{t5}}{P_{t4.5}} \right)^{\frac{(\gamma-1)}{\gamma}} \Rightarrow P_{t5} = P_{t4.5} \left(\frac{T_{t5 \text{ ideal}}}{T_{t4.5}} \right)^{\frac{\gamma}{\gamma-1}} = 392.01 \text{ kPa}$$

Off-design Point

$$\frac{T_{t5 \text{ ideal}}}{T_{t4.5}} = \left(\frac{P_{t5}}{P_{t4.5}} \right)^{\frac{(\gamma-1)}{\gamma}} \Rightarrow P_{t5} = P_{t4.5} \left(\frac{T_{t5 \text{ ideal}}}{T_{t4.5}} \right)^{\frac{\gamma}{\gamma-1}} = 122.68 \text{ kPa}$$

Turbine Expansion – Iteration Checks

Design point (Both turbines)

$$\left(\frac{\dot{m}_{4.5} \sqrt{T_{t4.5}}}{P_{t4.5}} \right)_{DP} = 8.10431$$

$$\left(\frac{\dot{m}_{4.5} \sqrt{T_{t4.5}}}{P_{t4.5}} \right)_{O-D} = 8.10428$$

Off-design point (Both turbines)

$$\left(\frac{\dot{m}_5 \sqrt{T_{t5}}}{P_{t5}} \right)_{DP} = 13.2021$$

$$\left(\frac{\dot{m}_5 \sqrt{T_{t5}}}{P_{t5}} \right)_{O-D} = 13.2024$$

$$\left(\frac{\dot{m}_{4.5} \sqrt{T_{t4.5}}}{P_{t4.5}} \right)_{ERROR} = \frac{\left(\frac{\dot{m}_{4.5} \sqrt{T_{t4.5}}}{P_{t4.5}} \right)_{O-D} - \left(\frac{\dot{m}_{4.5} \sqrt{T_{t4.5}}}{P_{t4.5}} \right)_{DP}}{\left(\frac{\dot{m}_{4.5} \sqrt{T_{t4.5}}}{P_{t4.5}} \right)_{DP}} = 0.000\%$$

$$\left(\frac{\dot{m}_5 \sqrt{T_{t5}}}{P_{t5}} \right)_{ERROR} = \frac{\left(\frac{\dot{m}_5 \sqrt{T_{t5}}}{P_{t5}} \right)_{O-D} - \left(\frac{\dot{m}_5 \sqrt{T_{t5}}}{P_{t5}} \right)_{DP}}{\left(\frac{\dot{m}_5 \sqrt{T_{t5}}}{P_{t5}} \right)_{DP}} = 0.002\%$$

Turbine expansion = constant

$$\frac{\left(\frac{\dot{m}_4 \sqrt{T_{t4}}}{P_{t4}} \right)_{DP}}{\left(\frac{\dot{m}_{4.5} \sqrt{T_{t4.5}}}{P_{t4.5}} \right)_{DP}} = \frac{\left(\frac{\dot{m}_4 \sqrt{T_{t4}}}{P_{t4}} \right)_{O-D}}{\left(\frac{\dot{m}_{4.5} \sqrt{T_{t4.5}}}{P_{t4.5}} \right)_{O-D}} \approx \frac{\sqrt{\tau_{HPT}}}{\pi_{HPT}} \approx constant$$

$$\frac{\left(\frac{\dot{m}_{4.5} \sqrt{T_{t4.5}}}{P_{t4.5}} \right)_{DP}}{\left(\frac{\dot{m}_5 \sqrt{T_{t5}}}{P_{t5}} \right)_{DP}} = \frac{\left(\frac{\dot{m}_{4.5} \sqrt{T_{t4.5}}}{P_{t4.5}} \right)_{O-D}}{\left(\frac{\dot{m}_5 \sqrt{T_{t5}}}{P_{t5}} \right)_{O-D}} \approx \frac{\sqrt{\tau_{LPT}}}{\pi_{LP}} \approx constant$$

Nozzle

Since the flow in the nozzle is adiabatic

$$\left(\frac{P_{t7}}{P_{t0}}\right)^*_{DP} = \left[\frac{\gamma + 1}{2}\right]^{\frac{\gamma}{\gamma-1}} = 1.851$$

$$\left(\frac{P_{t7}}{P_{t0}}\right)^*_{O-D} = \left[\frac{\gamma + 1}{2}\right]^{\frac{\gamma}{\gamma-1}} = 1.851$$

real > critical nozzle is choked

Design Point

$$P_9 = \frac{P_{t7}}{\left(\frac{P_{t7}}{P_{t0}}\right)^*} = 207.59 \text{ kPa}$$

$$T_9 = \frac{T_{t7}}{\left(\frac{T_{t7}}{T_{t0}}\right)^*} = 971.42 \text{ K}$$

$$T_{t7} = T_{t9}$$

$$\left(\frac{P_{t7}}{P_{t0}}\right)_{DP} = 3.791$$

$$\left(\frac{P_{t7}}{P_{t0}}\right)_{O-D} = 17.175$$

Off-design Point

$$P_9 = \frac{P_{t7}}{\left(\frac{P_{t7}}{P_{t0}}\right)^*} = 64.96 \text{ kPa}$$

$$T_9 = \frac{T_{t7}}{\left(\frac{T_{t7}}{T_{t0}}\right)^*} = 1050.18 \text{ K}$$

Thrust Specific Fuel Consumption

Design Point

Off-design Point

From static conditions in the nozzle, the speed of sound is calculated

$$\vec{V}_{\text{jet}} = \vec{V}_9 = \sqrt{\gamma_9 R_9 T_9} = 567.3 \text{ m/s}$$

$$\vec{V}_{\text{jet}} = \vec{V}_9 = \sqrt{\gamma_9 R_9 T_9} = 588.7 \text{ m/s}$$

Density is calculated from the pressure and temperature of the exhaust gases

$$\rho_9 = \frac{P_9}{T_9 R_9} = 0.745 \text{ kg/m}^3$$

$$\rho_9 = \frac{P_9}{T_9 R_9} = 0.222 \text{ kg/m}^3$$

Therefore area is as follows

$$A_9 = \frac{\dot{m}_4}{\rho_9 \vec{V}_9} = 0.36397 \text{ m}^2$$

$$A_9 = \frac{\dot{m}_4}{\rho_9 \vec{V}_9} = 0.36281 \text{ m}^2$$

Last Check!!
1.1 mm diff
Diameter within tolerance

Thrust is calculated as:

$$F_N = \dot{m}_9 (\vec{V}_{\text{jet}} - V_o) + A_9 (p_9 - p_o) = 125.956 \text{ kN}$$

$$F_N = \dot{m}_9 (\vec{V}_{\text{jet}} - V_o) + A_9 (p_9 - p_o) = 24.124 \text{ kN}$$

Specific fuel consumption

$$SFC = \frac{\dot{m}_f}{F_N} = 109.84 \frac{\text{kg/h}}{\text{kN}}$$

$$SFC = \frac{\dot{m}_f}{F_N} = 179.94 \frac{\text{kg/h}}{\text{kN}}$$

Engine Off-design Calculation

Turbojet - Mattingly

Performance analysis variables for single-spool turbojet engine

Component	Independent Variable	Constant or known quantity	Dependent Variable
Engine	M_0, T_{t0}, P_{t0}		$\dot{m}_0$
Diffuser		$\pi_d = f(M_0)$	
Low-pressure compressor		η_c	π_c, τ_c
Burner	T_{t4}	π_b, η_b	f
Low-pressure turbine		π_t, τ_t	
Exhaust nozzle	p_0/p_9	π_n	$M_9, T_9/T_0$
Total Number	5		9

Example 8.3:

We consider the performance of the turbojet engine described with the following data at the reference condition and altitude of 12 km.

Determine the engine's performance at an altitude of 9 km, Mach number of 1.5, reduced throttle setting ($T_{t4} = 1670$ K) and exit to ambient pressure ratio of $p_0/p_9 = 0.955$.

Design Point Conditions

Flight conditions			
Static			
Temperature	C		-56
Pressure	kPa		19.400
M0	-		2
Design			
Pressure Ratios			
Inlet	-		0.879
Compressor	-		10
Burner	-		0.94
Nozzle	-		0.96
Efficiencies			
Compressor Polytropic	-		0.9
Burner	-		0.98
Turbine Polytropic			0.9
Shaft/Mechanical			0.99
TIT	K		1800
Air mass flow rate	kg/s		50
p_0/p_9			0.5

Off-design Point Conditions

Flight conditions			
Static			
Temperature	C		- 43
Pressure	kPa		30.800
M0	-		1.8
Design			
Pressure Ratios			
Intake Pressure Recovery			0.922
TIT	K		1670
p_0/p_9			0.955
Gas Properties			
Cp Cold Section	J/kg.K		1004
γ cold	-		1.4
Cp Hot Section	J/kg.K		1239
γ hot	-		1.3
FHV	kJ/kg		42.800

Inlet

Design Point

$$T_{t0} = 390.1 \text{ K}$$

$$P_{t0} = 151.794 \text{ kPa}$$

$$a_0 = 295 \text{ m/s} ; V_0 = 590 \text{ m/s};$$

$$P_{t2} = P_{t0} \pi_d = 151.8 \cdot 0.879 \\ = 133.389 \text{ kPa}$$

Off Design

$$T_{t0} = 333.2 \text{ K}$$

$$P_{t0} = 113.068 \text{ kPa}$$

$$a_0 = 303.8 \text{ m/s} ; V_0 = 455.7 \text{ m/s};$$

$$P_{t2} = P_{t0} \pi_d = 104.254 \text{ kPa} \cdot 0.922 \\ = 104.254 \text{ kPa}$$

The analysis for the off – design starts just as the design conditions, assuming an adiabatic inlet.

The inlet discharge pressure is the pressure times the pressure ratio (P_{exit}/P_{inlet}).

LP Compressor

Design Point

$$T_{t2} = 390.1 \text{ K}$$

$$P_{t2} = 133.389 \text{ kPa}$$

Off Design

$$T_{t2} = 333.2 \text{ K}$$

$$P_{t2} = 104.254 \text{ kPa}$$

Compressor Inlet Conditions:

Design Point

From the design point compressor pressure ratio and the polytropic efficiency we can calculate the temperature ratio

$$P_{t3} = P_{t2} \pi = 133.4 \text{ kPa} \cdot 10 = 1333.893 \text{ kPa}$$

$$T_{t3} = T_{t2} \left(\frac{P_{t3}}{P_{t2}} \right)^{\frac{\gamma-1}{\gamma \eta_c}} = 810.2 \text{ K}$$

Compressor isentropic efficiency is calculated as follows:

$$\eta_c = \frac{\left(\frac{P_{t3}}{P_{t2}} \right)^{\frac{\gamma-1}{\gamma}} - 1}{\left(\frac{T_{t3}}{T_{t2}} \right) - 1} = 0.8641 = 86.4\%$$

Off-design Point

Using the temperature ratio from the design point, the off-design temperature ratio will be calculated as follows

$$\left(\frac{T_{t3}}{T_{t2}} \right)_{OD} = 1 + \left(\left(\frac{T_{t3}}{T_{t2}} \right)_D - 1 \right) \frac{(T_{t4}/T_{t2})_{OD}}{(T_{t4}/T_{t2})_D} = 2.1731 \therefore T_{t3} = 724.1 \text{ K}$$

With the previous temperature ratio and compressor efficiency of the off-design point, we can calculate the off-design compressor pressure ratio

$$\pi_c = \left[1 + \eta_c \left(\left(\frac{T_{t3}}{T_{t2}} \right) - 1 \right) \right]^{\frac{\gamma_c-1}{\gamma_c}} = 11.529 \therefore P_{t3} = 1201.947 \text{ kPa}$$

Combustor

Design point

Combustor efficiency is a given value. Therefore, the fuel mass flow can be calculated.

$$\eta_{cc} = \frac{(\dot{m}_0 + \dot{m}_f)h_4 - \dot{m}_{aire}h_3}{\dot{m}_f(LHV)} \left(\frac{\Delta h_{real}}{\Delta h_{ideal}} \right)$$
$$\Rightarrow \frac{\dot{m}_f}{\dot{m}_0} = \frac{(h_4 - h_3)}{(\eta_{cc} \cdot LHV - h_4)} = \frac{(1.239 \cdot 1800 - 1.004 \cdot 810.2)}{(0.98 \cdot 42,800 - 1.239 \cdot 1800)} = 0.03567$$

Off-design point

Same procedure is followed to calculate the fuel mass flow rate, using the same efficiency for the design point calculation

$$\eta_{cc} = \frac{(\dot{m}_0 + \dot{m}_f)h_4 - \dot{m}_0h_3}{\dot{m}_f(LHV)} \left(\frac{\Delta h_{real}}{\Delta h_{ideal}} \right)$$
$$\Rightarrow \frac{\dot{m}_f}{\dot{m}_0} = \frac{(h_4 - h_3)}{(\eta_{cc} \cdot LHV - h_4)} = \frac{(1.239 \cdot 1670 - 1.004 \cdot 724.1)}{(0.98 \cdot 42,800 - 1.239 \cdot 1670)} = 0.03366$$

Turbine

Turbine inlet conditions are given from design characteristics:

Design Point

$$T_{t4} = 1800 \text{ K}$$

$$P_{t4} = 1253.86 \text{ kPa}$$

Off Design

$$T_{t4} = 1670 \text{ K}$$

$$P_{t4} = 1129.83 \text{ kPa}$$

From power balance we obtain temperature at turbine exit:

$$T_{t5} = T_{t4} - \frac{C_{pC}(T_{t3} - T_{t2})}{\eta_M C_{pH}(1 + f)} = 1467.95 \text{ K}$$

$$T_{t5} = T_{t4} - \frac{C_{pC}(T_{t3} - T_{t2})}{\eta_M C_{pH}(1 + f)} = 1360.47 \text{ K}$$

Design Point

$$\frac{T_{t5}}{T_{t4}} = \left(\frac{P_{t5}}{P_{t4}} \right)^{\frac{(\gamma-1) \cdot e_t}{\gamma}} \Rightarrow P_{t4.5} = P_{t4} \left(\frac{T_{t5}}{T_{t4}} \right)^{\frac{\gamma}{(\gamma-1)e_t}} = 469.731 \text{ kPa}$$

Off-design Point

$$\frac{T_{t5 \text{ ideal}}}{T_{t4}} = \left(\frac{P_{t5}}{P_{t4}} \right)^{\frac{\gamma}{\gamma-1}} \Rightarrow P_{t5} = P_{t4} \left(\frac{T_{t5 \text{ ideal}}}{T_{t4}} \right)^{\frac{\gamma}{(\gamma-1)e_t}} = 421.083 \text{ kPa}$$

Nozzle

Since the flow in the nozzle is adiabatic

$$T_{t5} = T_{t7} = T_{t9}$$

From the design criteria both total and static pressure can be established, for design point would be as follows:

$$P_{t9} = \pi_n P_{t7} = 0.96(469.731) = 450.942 \text{ kPa and}$$

$$P_9 = \frac{P_0}{P_0/P_9} = 19.400/0.5 = 38.800 \text{ kPa}$$

For the off design would be:

$$P_{t9} = \pi_n P_{t7} = 0.96(421.083) = 404.239 \text{ kPa and}$$

$$P_9 = \frac{P_0}{P_0/P_9} = 30.800/0.955 = 32.251 \text{ kPa}$$

Since $P_{t9}/P_9 \gg 2$ and real > critical the nozzle is choked

Design Point

$$M_9 = \sqrt{\frac{2}{\gamma_t - 1} [(P_{t9}/P_9)^{(\gamma_t - 1)/\gamma_t} - 1]}$$

$$M_9 = \sqrt{\frac{2}{1.3 - 1} [(11.622)^{(1.3 - 1)/1.3} - 1]} = 2.253$$

Off-design Point

$$M_9 = \sqrt{\frac{2}{\gamma_t - 1} [(P_{t9}/P_9)^{(\gamma_t - 1)/\gamma_t} - 1]}$$

$$M_9 = \sqrt{\frac{2}{1.3 - 1} [(12.534)^{(1.3 - 1)/1.3} - 1]} = 2.298$$

Air Mass Flow Rate

For the air mass flow rate, in terms of design point vs off design, the use of all parameters in the inlet is used

$$\dot{m}_{0,OD} = \dot{m}_{0,D} \frac{(P_{t0}\pi_d\pi_c)_{OD}}{(P_{t0}\pi_d\pi_c)_D} \sqrt{\frac{T_{t4D}}{T_{t4OD}}} = \dot{m}_{0,D} \frac{P_{t3OD}}{P_{t3D}} \sqrt{\frac{T_{t4D}}{T_{t4OD}}}$$

$$\dot{m}_{0,OD} = 50 \frac{1201.947}{1333.893} \sqrt{\frac{1800}{1670}} = 46.775 \frac{kg}{s}$$

Performance Parameters

Design Point

From static conditions in the nozzle, the speed of sound is calculated

$$\vec{V}_{\text{jet}} = \vec{V}_9 = \sqrt{\gamma_9 R_9 T_9} = 1253.9 \text{ m/s}$$

Density is calculated from the pressure and temperature of the exhaust gases

$$\rho_9 = \frac{P_9}{T_9 R_9} = 0.163 \text{ kg/m}^3$$

Therefore, area is as follows

$$A_9 = \frac{\dot{m}_4}{\rho_9 \vec{V}_9} = 0.2536 \text{ m}^2$$

Thrust is calculated as:

$$F_N = \dot{m}_9 (\vec{V}_{\text{jet}} - V_o) + A_9 (p_9 - p_o) = 39.300 \text{ kN}$$

Specific fuel consumption

$$SFC = \frac{\dot{m}_f}{F_N} = 163.39 \frac{\text{kg/h}}{\text{kN}}$$

Off-design Point

$$\vec{V}_{\text{jet}} = \vec{V}_9 = \sqrt{\gamma_9 R_9 T_9} = 588.7 \text{ m/s}$$

$$\rho_9 = \frac{P_9}{T_9 R_9} = 0.222 \text{ kg/m}^3$$

$$A_9 = \frac{\dot{m}_4}{\rho_9 \vec{V}_9} = 0.2665 \text{ m}^2$$

$$F_N = \dot{m}_9 (\vec{V}_{\text{jet}} - V_o) + A_9 (p_9 - p_o) = 37.377 \text{ kN}$$

$$SFC = \frac{\dot{m}_f}{F_N} = 151.63 \frac{\text{kg/h}}{\text{kN}}$$

Performance analysis variables for dual-spool afterburner turbojet engine

Component	Independent Variable	Constant or known quantity	Dependent Variable
Engine	M_0, T_{t0}, P_{t0}	$\dot{m}_0$	
Diffuser		$\pi_d = f(M_0)$	
Low-pressure compressor		η_{cL}	π_{cL}, τ_{cL}
High-Pressure compressor		η_{cH}	π_{cH}, τ_{cH}
Burner	T_{t4}	π_b, η_b	f
High-pressure turbine		π_{tH}, τ_{tH}	
Low-pressure turbine		π_{tL}, τ_{tL}	
Afterburner	T_{t7}	π_{AB}, η_{AB}	f_{AB}
Exhaust nozzle	p_0/p_9	π_n	$M_9, T_9/T_0$
Total Number	6		9

Performance analysis variables for dual-spool turbofan with mixed exhaust

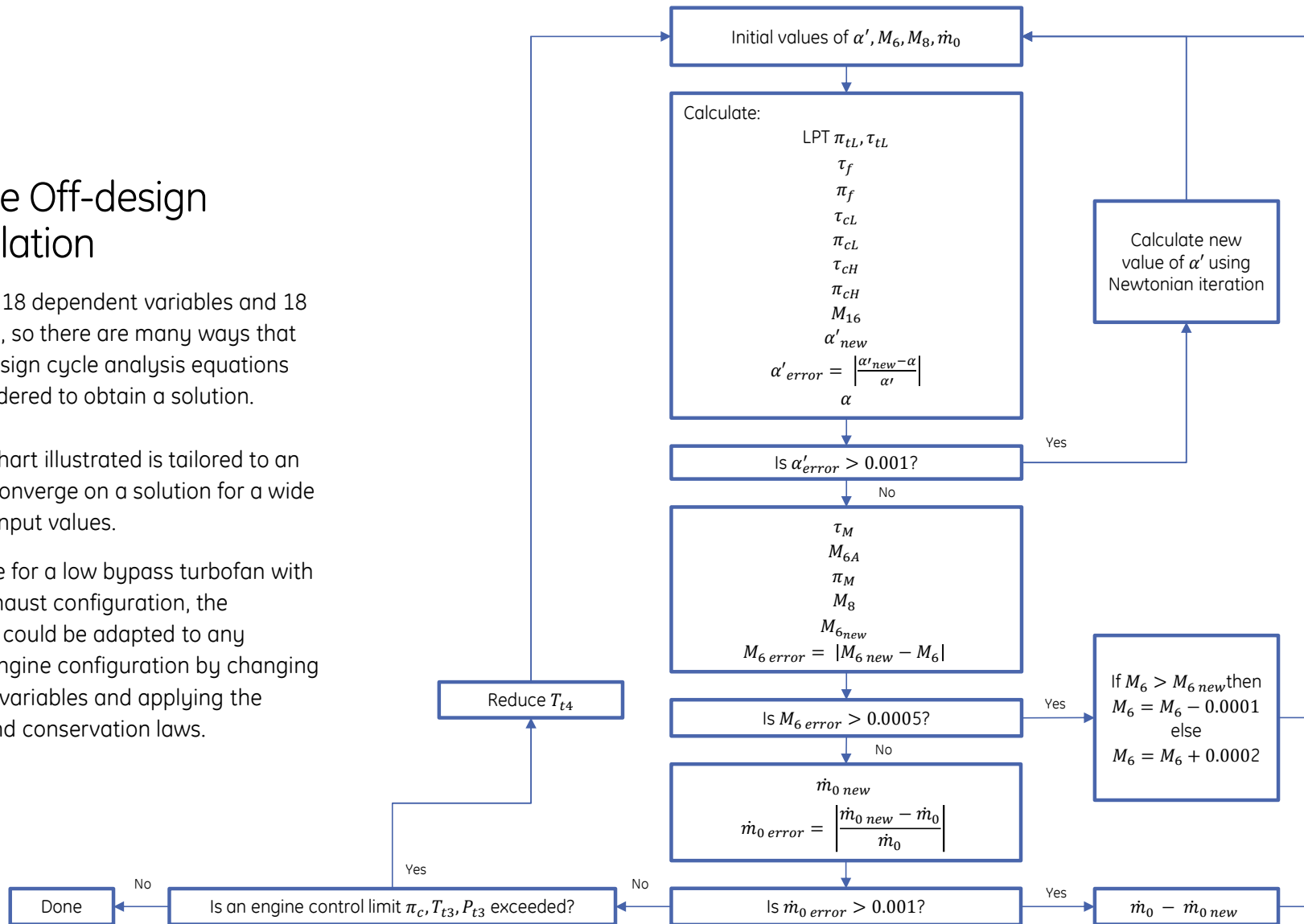
Component	Independent Variable	Constant or known quantity	Dependent Variable
Engine	M_0, T_{t0}, P_{t0} T_{t4} T_{t7}	β	$\dot{m}_0, \alpha$
Diffuser		$\pi_d = f(M_0)$	
Fan		η_f	π_f, τ_f
Low-pressure compressor		η_{cL}	π_{cL}, τ_{cL}
High-Pressure compressor		η_{cH}	π_{cH}, τ_{cH}
Burner		π_b	
Coolant mixer 1		ε_1	
High-pressure turbine		η_{tH}, M_4	
Coolant mixer 2		ε_2	
Low-pressure turbine		$\eta_{tL}, M_{4.5}$	π_{tL}, τ_{tL}
Mixer	p_0/p_9	$\pi_{Mmax}, A_6, A_{16}, A_{6A}$	$\pi_M, \tau_M, \alpha', M_6, M_{16}, M_{6A}$
Afterburner		$\pi_{AB \text{ dry}}, \pi_{AB}$	
Exhaust nozzle		$\pi_n, A_{8 \text{ dry}}$	M_8, M_9
Total Number	6		18

Engine Off-design Calculation

There are 18 dependent variables and 18 equations, so there are many ways that the off-design cycle analysis equations can be ordered to obtain a solution.

The flowchart illustrated is tailored to an that will converge on a solution for a wide range of input values.

Applicable for a low bypass turbofan with mixed exhaust configuration, the flowchart could be adapted to any existing engine configuration by changing the initial variables and applying the energy and conservation laws.



Engine Off-design Calculation

Power Conservation - Farokhi

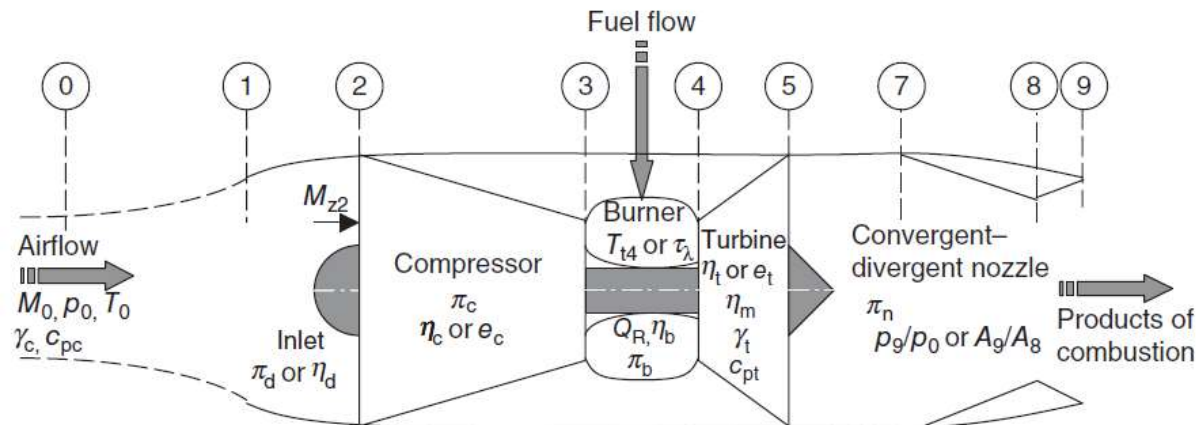
Engine Off-design Calculation

- As in design point calculations, the off-design analysis is a component step calculation procedure. Using the available data we must calculate the mass and energy changes across the engine to determine the behavior of the components. The known set design parameters is:

1. Design flight condition		M_0, p_0, T_0
2. Design compressor pressure ratio	π_c	
3. Ideal heating value of the fuel		Q_R
4. Design turbine inlet temperature		T_{t4} (or expressed as τ_λ)
5. Design component efficiencies		$\pi_d, e_c, \pi_b, \eta_b, e_t, \eta_m, \pi_n$
6. Design nozzle exit pressure or area ratio		p_9/p_0 or A_9/A_8
7. Gas properties at design point		$\gamma_c, C p_c, \gamma_t, C p_t$
8. Design corrected mass flow rate		$\dot{m}_{c2}$
9. Design corrected shaft speed		N_{c2}
10. Design axial Mach number at 2		M_{z2}

Engine Off-design Calculation

- Then the engine is subject to off-design conditions, namely,
 - Off-design flight conditions (specified) M_0, p_0, T_0
 - Off-design turbine inlet temperature (specified) T_{t4} or τ_λ
- Assumptions:
 - Constant component efficiencies
 - Only changing parameter is pressure ratio
 - Pressure ratio establishes corrected mass flow rate and corrected shaft speed



Example 11.2:

Assume a turbojet engine with the parameters as shown in table 1 for design point and a perfectly expanded nozzle. If this engine is operating as specified in table 2 conditions and assuming all other component efficiencies remain the same at off-design calculate:

- $\pi_{C,OD}$
- $\dot{m}_{C2,OD}$
- $N_{C2,OD}$
- $M_{Z2,OD}$

Design Point

Flight conditions		
Static		
Temperature	C	15
Pressure	kPa	100
M_0	-	0
Gas Properties		
Cp Cold Section	J/kg.K	1004
γ cold	-	1.4
Cp Hot Section	J/kg.K	1156
γ hot	-	1.33
FHV	kJ/kg	42.000
Design		
Pressure Ratios		
Diffuser	-	0.98
Compressor	-	25
Burner	-	0.98
Nozzle		0.97
Efficiencies		
Compressor Polytropic	-	0.9
Burner	-	0.99
Turbine Polytropic		0.85
Shaft/Mechanical		0.995
TIT	C	1500
$\dot{m}_{C2}$	kg/s	73
M_{Z2}		0.6
N_{C2}	rpm	6000

Off-design Point

Flight conditions		
Static		
Temperature	C	-15
Pressure	kPa	33
M_0	-	0.8
Design		
Pressure Ratios		
Diffuser	-	0.995
TIT		
TIT	C	1375

Solution

SOLUTION

Power balance between the compressor and turbine at design point yields

$$T_{t2}(\tau_c - 1) = (1 + f)\eta_m T_{t4}(1 - \tau_t)$$

Therefore, we may isolate the turbine expansion ratio τ_t to get

$$\tau_t = 1 - \frac{1}{(1 + f)\eta_m} \frac{T_{t2}}{T_{t4}} (\tau_c - 1) = 1 - \frac{1}{(1 + f)\eta_m} \frac{\tau_r}{\tau_\lambda} (\tau_c - 1)$$

The same turbine expansion τ_t between the on- and off-design operations (as derived earlier), as well as assuming nearly constant $(1 + f)\eta_m$ between the on- and off-design modes, suggest that

$$\left. \frac{\tau_r}{\tau_\lambda} (\tau_c - 1) \right|_{\text{off-design}} \approx \left. \frac{\tau_r}{\tau_\lambda} (\tau_c - 1) \right|_{\text{design}}$$

The off-design compressor temperature ratio is

$$\tau_{c,\text{O-D}} \approx 1 + \frac{(\tau_t/\tau_\lambda)_D}{(\tau_t/\tau_\lambda)_{\text{O-D}}} (\tau_c - 1)_D$$

Solution

Now, let us calculate the parameters and substitute in the above equation

$$(\tau_r/\tau_\lambda)_D = \left(\frac{T_{t2}}{T_{t4}}\right)_D = \left(\frac{T_0}{T_{t4}}\right)_D = \frac{273 + 15}{273 + 1500} \cong 0.16244$$

$$(\tau_c)_D = (\pi_c)_D^{\frac{\gamma-1}{\gamma}} = (25)^{\frac{0.4}{(0.9)(1.4)}} \cong 2.7784$$

$$(\tau_r/\tau_\lambda)_{O-D} = \left(\frac{T_{t2}}{T_{t4}}\right)_{O-D} = \left(\frac{T_0[1 + (\gamma - 1)M_0^2/2]}{T_{t4}}\right)_{O-D} = \frac{(273 - 15)[1 + 0.2(0.64)]}{273 + 1375} \cong 0.1766$$

Therefore,

$$\tau_{c,O-D} \approx 1 + \frac{0.1624}{0.1766}(2.7784 - 1) \cong 2.6354$$

Hence, the compressor pressure ratio at off-design is

$$\pi_{c,O-D} = [(\tau_c)_{O-D}]^{\frac{\gamma}{\gamma-1}} = (2.6354)^{0.9(3.5)} \cong 21.2 \quad \Rightarrow$$

Off-design $\pi_c \approx 21.2$

Solution

With the compressor pressure ratio calculated at off-design, we may proceed to calculate engine performance parameters and figures of merit, such as specific thrust, specific fuel consumption, and thermal and propulsive efficiencies. The corrected mass flow rate at off-design is related to the design value by Equation 11.31.

$$\frac{\dot{m}_{c2,O-D}}{\dot{m}_{c2,D}} \cong \frac{\pi_{c,O-D}}{\pi_{c,D}} \frac{\sqrt{(T_{t4}/T_{t2})_D}}{\sqrt{(T_{t4}/T_{t2})_{O-D}}} = \frac{21.2}{25} \sqrt{\frac{0.1766}{0.16244}} \approx 0.8842$$

$$\dot{m}_{c2,O-D} \approx 0.8842(73 \text{ kg/s}) \approx 64.5 \text{ kg/s}$$



Off-design $\dot{m}_{c2} \approx 64.5 \text{ kg/s}$
--

The off-design corrected shaft speed is related to the design value via Equation 11.28b,

$$\frac{N_{c2,O-D}}{N_{c2,D}} = \frac{\sqrt{(T_{t4}/T_{t2})_{O-D}}}{\sqrt{(T_{t4}/T_{t2})_D}} = \sqrt{\frac{0.16244}{0.1766}} \approx 0.959$$

$$N_{c2,O-D} \text{ (rpm)} \approx 0.9591 (6000) \approx 5754$$



Off-design $N_{c2} \approx 5754 \text{ rpm}$

Solution

In addition to the corrected mass flow rate at off-design, we can calculate the physical mass flow rate at off-design as well. For that purpose, Equation 8.10 may be written as

$$\dot{m}_2 = \dot{m}_{c2} \delta_2 / \sqrt{\theta_2}$$
$$\delta_2 \equiv p_{t2} / p_{\text{ref}} \text{ where } p_{\text{ref}} = 101.33 \text{ kPa}$$

$$p_{t0} = p_0 (1 + 0.2 M_0^2)^{3.5} = 33 \text{ kPa} [1 + 0.2(0.64)]^{3.5} \approx 50.3 \text{ kPa}$$

$$p_{t2} = \pi_d p_{t0} = (0.995)(50.3 \text{ kPa}) = 50.05 \text{ kPa}, \text{ therefore } \delta_2 = 50.05 \text{ kPa} / 101.33 \text{ kPa} = 0.4939$$

$$\theta_2 \equiv T_{t2} / T_{\text{ref}} \text{ where } T_{\text{ref}} = 288.2 \text{ K}$$

$$T_{t2} = T_0 (1 + 0.2 M_0^2) = 258 \text{ K} [1 + 0.2(0.64)] \approx 291 \text{ K}, \text{ therefore } \theta_2 = 291 \text{ K} / 288.2 \text{ K} \approx 1.01$$

We may substitute these values in the mass flow rate equation, to get

$$\dot{m}_2 = \dot{m}_{c2} \delta_2 / \sqrt{\theta_2} = 64.5 \text{ kg/s} (0.4939) / \sqrt{1.01} \approx 31.7 \text{ kg/s} \quad \Rightarrow$$

$\frac{\text{Off-design}}{\dot{m}_2 \approx 31.7 \text{ kg/s}}$

Solution

Now, we can calculate the off-design axial Mach number at the engine face, M_{z2} .

From the definition of corrected mass flow rate

$$\dot{m}_{c2} = \frac{\dot{m}_2 \sqrt{\theta_2}}{\delta_2} = \sqrt{\frac{\gamma_2}{R_2}} \frac{p_{\text{ref}}}{\sqrt{T_{\text{ref}}}} \cdot A_2 \cdot M_{z2} \left(1 + \frac{\gamma_2 - 1}{2} M_{z2}^2 \right)^{\frac{-(\gamma_2 + 1)}{2(\gamma_2 - 1)}} = f(\gamma_2, R_2, M_{z2})$$

we conclude that

$$\frac{\dot{m}_{c2,D}}{\dot{m}_{c2,O-D}} = \frac{M_{c2,D}}{M_{c2,O-D}} \left[\frac{1 + (\gamma_2 - 1)M_{z2,D}^2/2}{1 + (\gamma_2 - 1)M_{z2,O-D}^2/2} \right]^{\frac{-(\gamma_2 + 1)}{2(\gamma_2 - 1)}} = \frac{73}{64.5} = \frac{0.6}{M_{z2,O-D}} \left[\frac{1 + 0.2M_{z2,O-D}^2}{1 + 0.2(0.6)^2} \right]^3$$

Solving for the only unknown in the above equation yields

$$M_{z2,O-D} = 0.4975 \quad \Rightarrow$$

Off-design $M_{z2} \approx 0.4975$

Note that although the physical mass flow rate dropped by more than half, the axial Mach number at the engine face dropped from 0.6 to ~0.5.